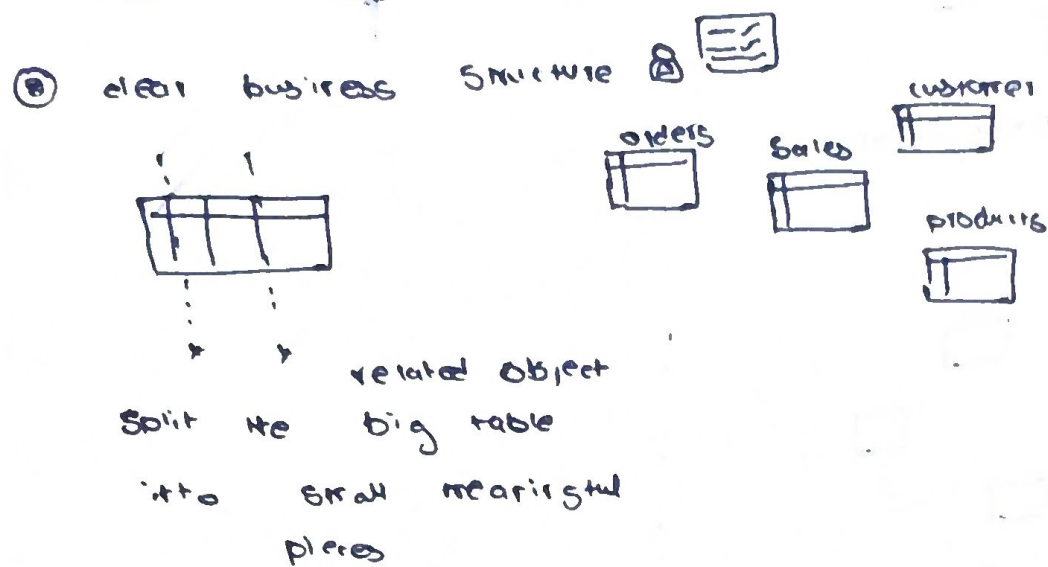
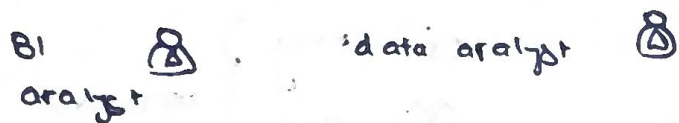


11 Sept 26

# Data Modeling



③ Design for consumption

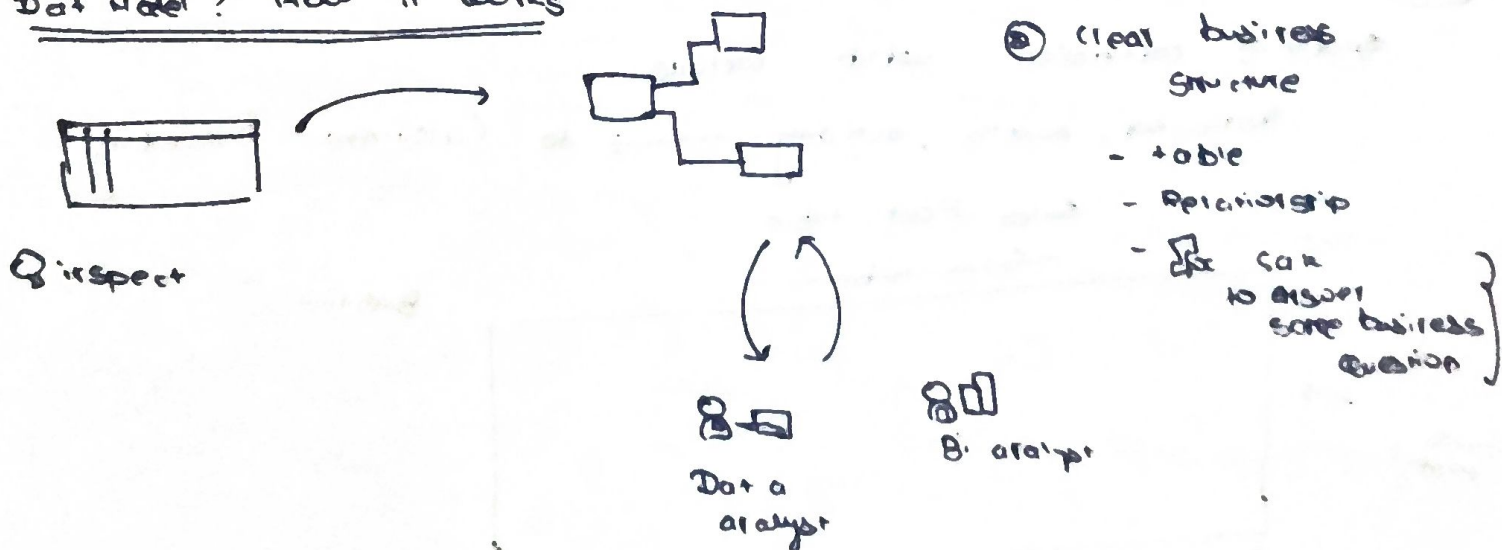


... Design Data model in a way  
so that BI, Data analyst work efficiently

## 3 components of Data model

- ① table
- ② Relationship :: how to connect the table
- ③ calculator :: to answer business question

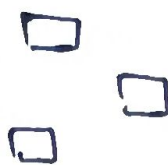
## Data Model? How it works



Tables :: Depends on technique table may take on  
Different role and shape



Big table

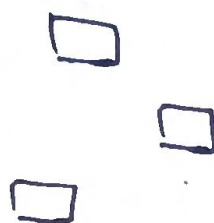


Group object

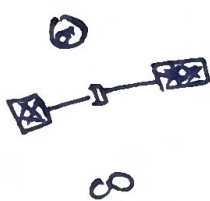
Data Modelling



Split information



Group object



Hub  
Satellite  
link  
Bridge

Data Vault



Entity



M:N

operation

For analytic & Reporting

- Dimensional modelling



Fact



Dimension

Facts

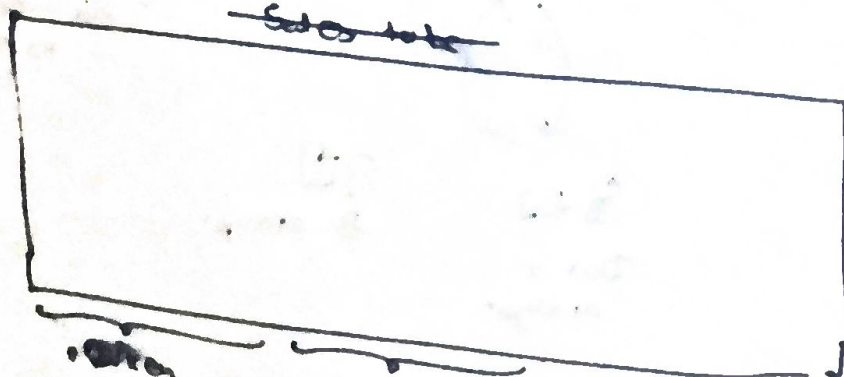
Quantitative information which contains

transaction, events, activity

→ to calculate something

Example Fact table

~~Sales table~~



Sales  
table?

Sales

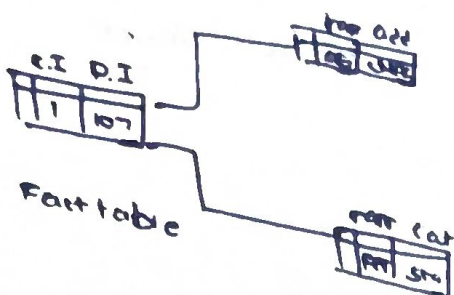
table

number

Tables

# Dimension

gives more context to the event in the fact



- contains descriptive info

- when connected to the event in fact it gives more context to the event

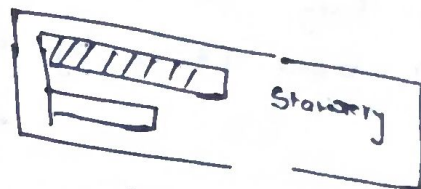
Describe something

| PI | P-Name | P-Color | P-Brand |
|----|--------|---------|---------|
| 01 | Red    | Black   |         |
|    |        |         |         |
|    |        |         |         |

1 Row = 1 Product

No event

Attributes (more context)



aggregation

EDF --- Dimension Exploration

Dimension

Describe "context"

Fact

What happened?

Attributes (labels)

key + time + number // core

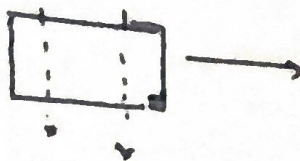
Sales by category



who was where when

What happened?

--- Revision til now



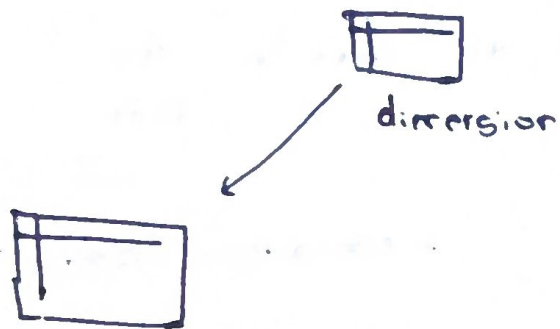
group order



D. H. technique  
↳ D. Matching

Sales by category

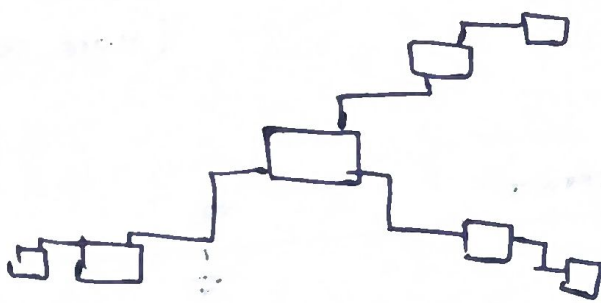
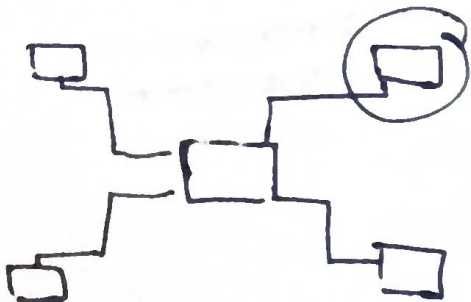
# Star Schema



Fact table  
Bigger in size

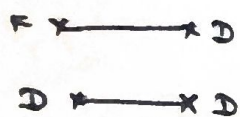
- ✓ Clean Filter
- ✓ Easy Data
- ✓ Visual (Sales by category)

Star Schema



In Real project ; we  
dimension will keep getting  
bigger

Split into  
multiple dimension



- Easy
- Fast
- Big size

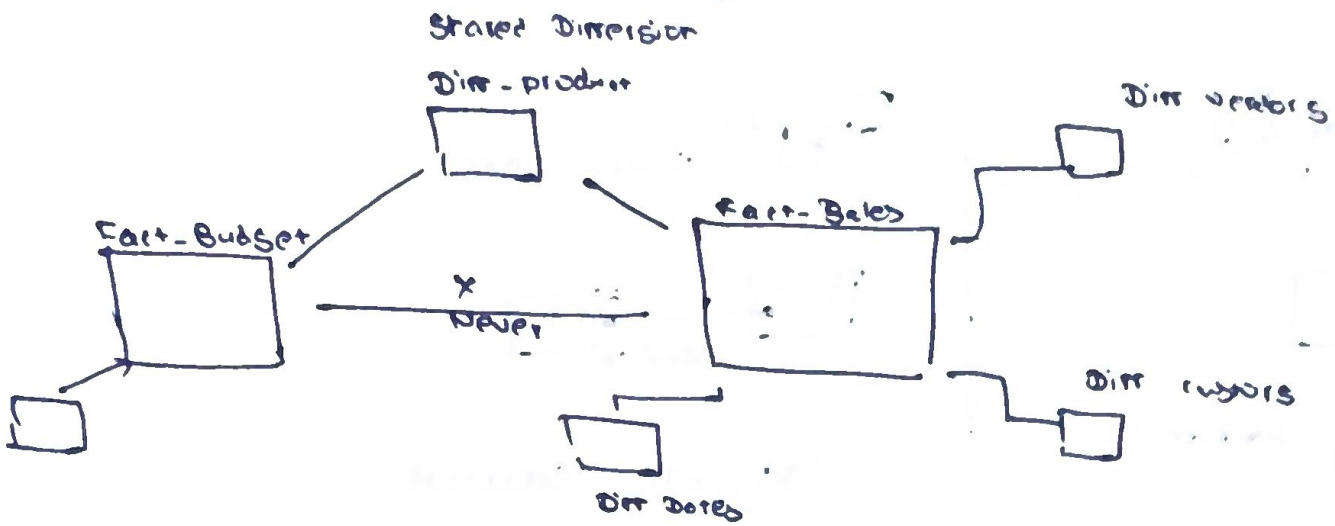
- complex
- (too many joins) slow
- ~~Small~~ Size

Default ✓

Avoid ✗

10/19/26

Problem this time is Fact (ore, perfect fact 0?)



Galaxy schema

Fact — Fact  
S.D

Dataset = Semantic model = Data Model



PowerBI

any platform



Data model

BTS

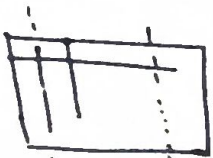


Report  
visual layer

5th layer

Business needs  
Table ♀ ♂

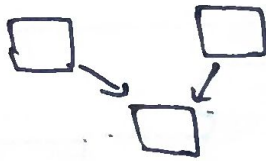
Dimension



Big table



group object



Dimension  
Fact

Dimensional  
technique

Star schema

Star schema

Galaxy schema



"Point"

"Big Dimension"

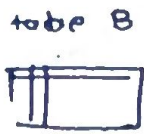
"Shared Dimension"

Data model

① table

② Relationship

Core : Structure → roles → Shape  
Idm



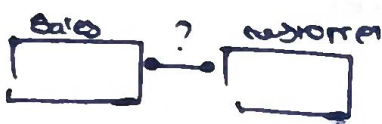
power Query



physically combine

physically separated

Problem

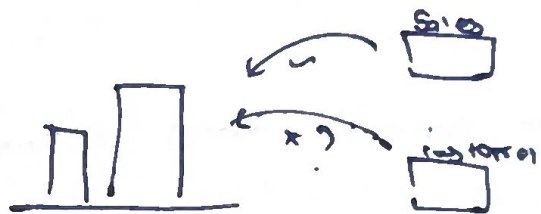


By defining the relation between two table



this is how you do it ! bro

visual layer



How to combine?

combine Data → Need Relation

Fields



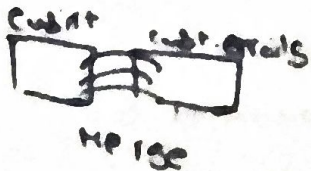
visual layer



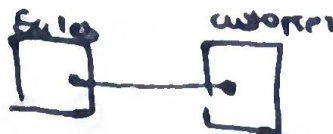
Model layer

physical layer

combination occurs here



Same entities



Relation

Great example for Multiple entities

1  
unique

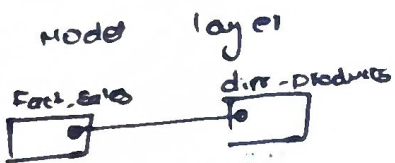
max

Eg:

Dim - customer

Dim - loyalty  
here

Revision-1 xx — xx

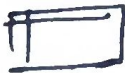


(AND)

Visual layer



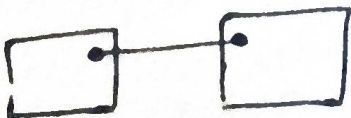
Model layer



Physical layer



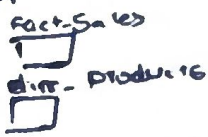
Base entities



tip  
[evil non]

Fact - order

Visual layer



dump on top of it

200m at -150%



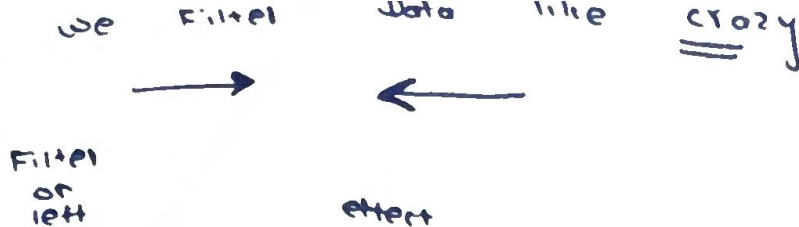
different entities

zoom in +150%

Key

Cardinality (very word)

Filtering

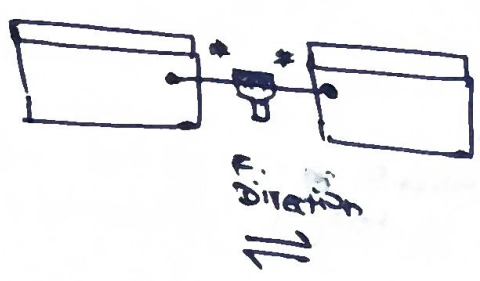


event ~~X~~ filtering on right

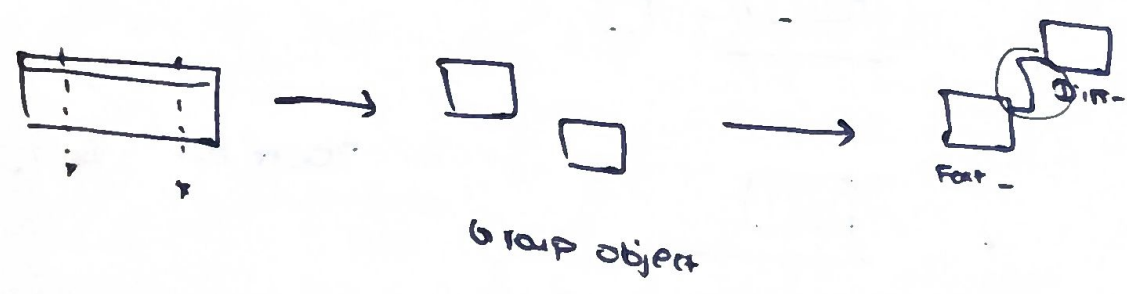
↳ control the whole behaviour of Data Model including filter  
I'm in charge now!

↳ good Report

Revisor - 2 :

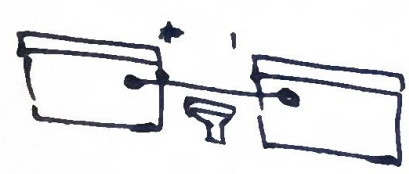


Revisor - 3 - \*\* - \*\* -



11 BI & Analytics

zoom in + 50%



- ① key
- ② cardinality
- ③ filter

Model layer



PowerBI  
Say that again?  
what

Visual layer  
Diff. customers